

Week 4 – End-to-End Data Analysis Capstone Solution

Dataset: Sample data (1).xlsx | 700 records | 16 original columns

1. EDA – Exploratory Data Analysis

The dataset contains 700 records and 16 original columns. Total Sales are 118,726,350.26, total Profit is 16,893,702.26, and total Units Sold are 1,125,806. Overall profit margin is 14.23%.

Country	Sales	Profit	Units
United States of America	25029830.16	2995540.66	232627.5
Canada	24887654.88	3529228.88	247428.5
France	24354172.28	3781020.78	240931.0
Germany	23505340.82	3680388.82	201494.0
Mexico	20949352.11	2907523.11	203325.0

2. Regression Model

A linear regression pipeline was created with one-hot encoding for categorical fields. The model predicts Sales using Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Month Number, Year and Month. Gross Sales, Discounts and COGS were excluded to avoid using direct components of the target variable.

Metric	Value
R ²	0.7643
MAE	82,876.5444
RMSE	114,244.4006

3. Power BI Dashboard

Recommended dashboard: KPI cards for Total Sales, Total Profit, Units Sold and Profit Margin; a monthly Sales/Profit trend; bar charts by Country, Segment and Product; and slicers for Year, Country, Segment, Product and Discount Band. The folder PowerBI_Data contains CSV files ready for import.

4. Findings & Recommendations

The dataset shows sales and profit variation across countries, products and customer segments. Germany has the highest sales total among the five countries in this supplied dataset. Missing Discount Band values were handled as Unknown. Use the dashboard to identify high-sales/high-profit combinations and areas where discounts coincide with weaker margins. Review the regression metrics before using the model for operational forecasting.

5. Deliverables

Excel solution, Python script, Jupyter notebook, Power BI-ready CSV data, and this PDF report are included. A .pbix file itself cannot be generated here, so the PowerBI_Data folder and dashboard specification are provided for direct Power BI import.